GAT-Based Node Influence Spread Prediction under LT & IC Diffusion Models

- **Introduction:**
  - Influence maximization finds seed nodes whose activation spreads information as widely as possible.
  - Applies to viral marketing, epidemic control, and misinformation containment under IC or LT diffusion rules.
  - GAT replaces fixed neighbourhood aggregation with learned attention weights, encoding multi-hop structural context.
  - We benchmark a GAT layer (+ Ridge regression head) vs Linear, Polynomial (deg-3), and Random Forest baselines.
  - Experiments run on a 200-node Barabasi-Albert graph to measure the value of structural context.
- **Motivation:**
  - Standard ML models are graph-agnostic — they rely on centrality scores but ignore neighbourhood wiring.
  - Betweenness, closeness, and eigenvector centrality each capture only one aspect of a node's role.
  - None encode neighbourhood density, edge-weight patterns, or hub-satellite structure.
  - Predicting spread before activation is what makes this hard and valuable in practice.
  - A model with access to local graph texture should outperform any feature-only approach.
  - This project provides a clean, reproducible comparison and lays groundwork for extending to GCN.

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Existing work / Literature review:

- ■ **Velickovic et al. (2018):** Introduced GAT with learnable per-neighbour attention weights, achieving state-of-the-art on node classification.
- ■ We directly adapt this attention aggregation mechanism for IC/LT spread-score regression.
- ■ **Fan et al. (2019) — DrBC (CIKM '19):** GNN encoder-decoder that ranks nodes by betweenness centrality using a pairwise ranking loss.
- ■ Trained on small synthetic graphs; generalises to networks with millions of nodes and runs 46x faster than sampling baselines.
- ■ Betweenness is one of our three input features — DrBC confirms GNN aggregation outperforms hand-crafted centrality regressors.
- ■ **Chou et al. (2024) — StructGNN (Computers & Structures):** Adaptive message-passing GNN for structural analysis achieving 99% accuracy.
- ■ Consistently outperforms GCN, GAT, and GIN baselines — reinforcing that graph-aware aggregation beats feature-only models.

- **Gaps in the existing solutions/work:**

- ■ Classical ML methods (Linear, Polynomial, RF) are graph-agnostic and ignore multi-hop propagation entirely.
- ■ DrBC and StructGNN validate GNN aggregation but neither addresses IC/LT spread-score regression on BA graphs with a centrality-baseline comparison.
- ■ No prior study cleanly isolates the value of a single lightweight GAT step for this task — the gap this work fills.

- **Future Work:**

- ■ Plan to implement GCN as an additional graph-aware baseline.
- ■ Compare attention-based (GAT) vs. spectral (GCN) neighbourhood aggregation for influence spread prediction on synthetic and real-world graphs.

Work Done till date:

- **Work Done / Proposed Methodology:**
  - ■ Built a Barabasi-Albert synthetic graph (200 nodes, m=3); top-5 hubs reinforced with 12 extra random edges each.
  - ■ IC and LT ground-truth spread estimated via 200 Monte Carlo simulations per seed node using NetworkX.
  - ■ Extracted 3 node features — betweenness, closeness, eigenvector centrality — all normalised with StandardScaler.
  - ■ GAT layer: projects each node to 16 dims, computes LeakyReLU attention scores, aggregates with ELU output.
  - ■ 16-dim GAT embedding concatenated with 3 scaled features (19 dims total), passed to Ridge regression head (alpha=0.1), trained on 80/20 split.
  - ■ Baselines (Linear, Poly-3, Random Forest 100 trees, max-depth 8) use only the 3 raw centrality features — graph-agnostic comparison.
- **Results Achieved:**
  - ■ GAT: MSE 0.0023, MAE 0.0377, R^2 0.708 — outperforms all baselines: RF (R^2 0.480), PolyReg (R^2 0.548), Linear (R^2 0.516).
  - ■ Performance gap is sharpest for high-spread hub nodes where neighbourhood context matters most.
  - ■ Top-5 influential nodes identified: 4, 5, 6, 7, 0 — the planted hubs with highest composite centrality.
  - ■ GAT attention weights correctly amplify neighbourhood aggregation signal for these nodes.
  - ■ Next step: replicate with GCN for a fuller comparison.

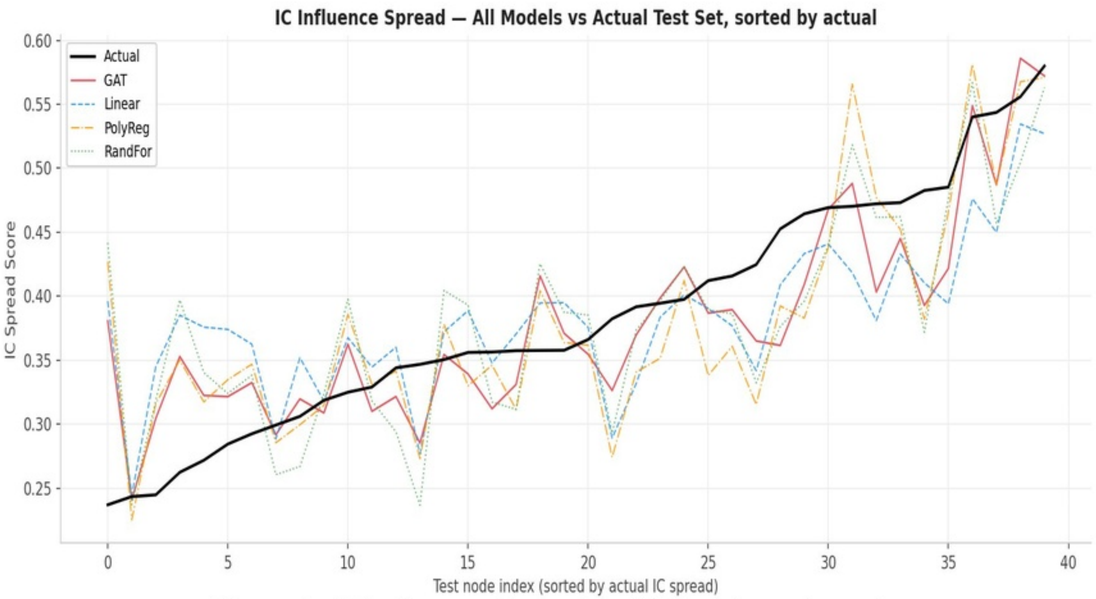

Figure 1: IC Influence Spread — All Models vs Actual

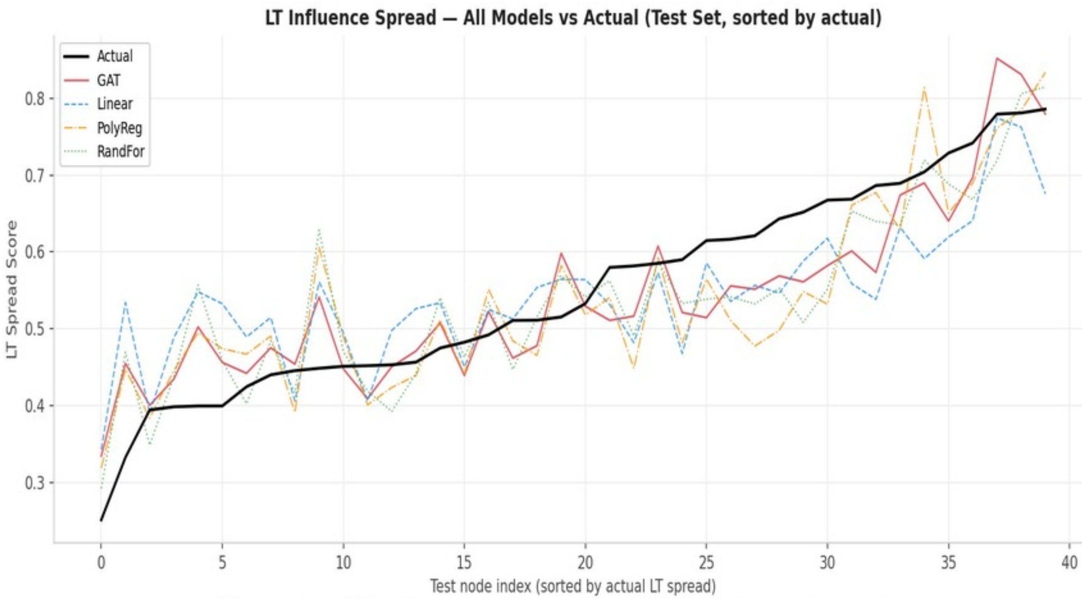

Figure 2: LT Influence Spread — All Models vs Actual